

HIKMAH *DataML*

Architecture & Operations — how this autonomous newsletter is built and run

Domain: Production Intelligence for ML Engineers

Auto: Thursday 06:00 GST

Vol. I · Global Edition

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01 What this does

HIKMAH DataML is a fully autonomous weekly intelligence newsletter. Every week a CrewAI agent crew reads 12 curated industry sources, removes anything already covered, scores what remains for a **senior audience** (engineers, architects, consultants, executives), and publishes the strongest **24 stories** — 6 in each of 4 sections — as desktop and mobile web pages and PDFs, archived on GitHub Pages.

It is not written for juniors: each story carries a decision-grade **"Signal"** that fuses the architectural implication with the strategic/business consequence — adoption timing, cost, risk, and the move a senior leader should make.

02 How CrewAI works here

CrewAI orchestrates a small team of role-specialised LLM agents that hand work to each other in a fixed sequence (a "crew"). Each agent has a role, a goal, a model, and optionally tools it can call. Output of one task becomes the context of the next.



03 The agents

#	AGENT	MODEL	TOOLS	RESPONSIBILITY
01	Data Scout	claude-haiku-4-5	fetch_rss, dedup	Fetches every RSS feed, removes already-seen articles via SHA-256 dedup (persisted in a per-project SQLite DB), and classifies the rest into the 4 sections.
02	Principal Analyst & Strategist	claude-sonnet-4-6	(reasoning only)	Scores each article 0-100 for senior-audience significance, drops anything below 65, and writes the 3-sentence summary plus the decision-grade 'Signal'. Keeps the top 6 per section.
03	Intelligence Publisher	claude-sonnet-4-6	(reasoning only)	Compiles the final, schema-valid JSON payload (issue metadata, stats, ticker, 4 sections x 6 entries) that drives the templates. Emits JSON only.

Models are addressed as anthropic/claude-* so CrewAI routes them to Anthropic. The Scout uses the fast Haiku model for high-volume triage; the Analyst and Publisher use Sonnet for judgement and strict formatting.

04 Context & data flow

- **Tools as context:** the Scout's `fetch_rss` tool returns raw articles; its `dedup` tool checks each URL's SHA-256 against a per-project SQLite DB so a story is never repeated across issues.
- **Task chaining:** each agent's output is passed as the *context* of the next task — Scout→Analyst→Publisher — so the Analyst sees the classified set and the Publisher sees the scored set.
- **Structured payload:** the Publisher emits one JSON object — issue metadata, stats, ticker headlines, and 4 sections x 6 entries (title, url, source, date, summary, signal, keywords, vendors, score).
- **Branding merged in code:** the project's colours, wordmark and section taxonomy (from `brand.py`) are merged into the payload before a shared Jinja2 template renders it.

05 Coverage — the 4 sections

01 · MLOps & Platforms

MLflow · Kubeflow · Feature Stores · Model Registry · CI/CD · Monitoring

Watching: Databricks · MLflow · dbt · Weights & Biases · Evidently AI

02 · ML & Deep Learning

PyTorch · Fine-Tuning · Transformers · Computer Vision · Time-Series

Watching: PyTorch · HuggingFace · scikit-learn · XGBoost · TensorFlow

03 · RAG, Vector & Data Infrastructure

RAG Architectures · Vector DBs · Embeddings · LangChain · LlamaIndex

Watching: Weaviate · Pinecone · Chroma · LangChain · Apache Iceberg

04 · Cloud ML: Azure & AWS

Azure ML · Prompt Flow · SageMaker · Bedrock · Vertex AI · Cost

Watching: Azure ML · SageMaker · Bedrock · Vertex AI · Cost Engineering

06 The suite it belongs to

This is one of **five independent projects** in the HIKMAH suite, each with its own crew, sources, dedup DB, brand and weekly slot, all sharing one library (`hikmah-shared`: `renderer`, `PDF generator`, `GitHub publisher`, `email sender`, `dedup`):

PROJECT	DOMAIN	DAY (06:00 GST)
signal	Telecom / 5G / RAN	Tuesday
intelligence	AI / Agentic / LLM	Wednesday
dataml	MLOps / Data Science	Thursday
cloudinfra	Cloud / Containers / Edge	Friday
dataarch	Databases / Big Data / GPU / API	Saturday

07 What an "issue" is

Each run produces one numbered issue with **4 deliverables** — published to `CrewAI/dataml/issues/` and served on GitHub Pages:

- `issue-NNN.html` — desktop edition (multi-column)
- `issue-NNN-mobile.html` — phone edition (single column)

- [issue-NNN.pdf](#) — A4 print/archive PDF
- [issue-NNN-mobile.pdf](#) — phone-format PDF

An issue number auto-increments each full run. Runs can be **manual** (`run_now.py dataml`) or **automatic** (the weekly scheduler), with an offline `--demo` mode for previewing without an API key.

08 Sources (12)

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| • huggingface.co | • netflixtechblog.com |
| • kdnuggets.com | • engineering.fb.com |
| • machinelearningmastery.com | • aws.amazon.com |
| • eugeneyan.com | • magazine.sebastianraschka.com |
| • towardsdatascience.com | • huyenchip.com |
| • databricks.com | • dataengineeringweekly.com |